

Sixteen Scientists on One Board

A society of piece-agents as a chess player, in which coordination is quorum sensing, strategy is a coalition formula, and a ply is a join

A F1R3FLY.io working note

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Abstract

This note specifies a chess player whose *player* is not an evaluator but a society: each piece of the moving colour is an agent in its own right, and the sixteen agents coordinate to emit one move per turn. The board is likewise an agent. The setting is cost-accounted ρ as implemented on the `cost-accounted-rho` branch of `f1r3node-rust`, with the generated spatial-behavioural logic of the mortal-scientist programme, and we assume at development time an implementation of coalition logic augmented with context-labelled modalities. We argue five structural points and then give a spec. First, the board needs three name families per square, not one: an occupancy cell, a move interface, and an attack-marker family, because perception must not be able to act and because the attack map is the one expensive object all sixteen agents share. Second, a chess ply is not a single rewrite but an N -ary *join*, and the join arity is exactly the move taxonomy: two for a quiet move, two with occupant termination for a capture, three for en passant, four for castling. This is what preserves the generated modality $\langle K_j \rangle$ and the bijection between redex positions and legal moves. Third, the coordination problem is the honeybee problem, and its solution is forced by the same altitude argument: consensus over sixteen pieces is a global universal with no presented witness, quorum at a candidate move is a local $\oplus \otimes$ over one open, so the society must quorum-sense. Fourth — and this is the load-bearing observation — in ρ a quorum *is* a join arity and one linear advocacy token per piece per round is the polity's franchise, so the two-quorum theorem becomes an arithmetic fact about sends rather than a simulation result: with sixteen pieces and threshold $k \geq 9$ a split is structurally impossible. Fifth, legality is a genuine universal and therefore belongs to the world, not to the society; pins and checks are deposited structure and pin-awareness is perception. We then observe that capture is predation in the existing sense, so material advantage is metabolic advantage and the classical piece values become something the ecology might derive rather than receive. The implementation spec that follows fixes the namespace discipline, the turn protocol, the formula ladder, the cost schedule, the requirements placed on the assumed coalition-logic checker, a six phase delivery plan, and eight measurements, of which the sharpest asks whether the noughts-and-crosses finding that non-modal rungs deliver ninety-two percent of yield for thirty-four percent of cost is a fact about games or a fact about noughts.

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1 Introduction

A conventional chess program is a single agent. It holds a position, generates moves, evaluates leaves, and returns an argmax. Everything about its architecture follows from that: the evaluation function is global, the search is centralised, and the pieces are data.

The proposal this note specifies inverts that. Each piece of the moving colour is an agent with its own budget, its own hypothesis about the position, and its own capacity to act. The board is an agent too. Sixteen agents want to move and exactly one move commits, so the architecture is obliged to contain something the conventional design never needs: a mechanism that turns sixteen concurrent desires into one commitment, without a chairman.

That obligation is the reason to build the thing. It is not a distributed implementation of a familiar algorithm. It is the smallest classical domain in which the constructions of the mortal-scientist programme — cost-accounted computation, the generated spatial-behavioural logic, the assay in guard position, predation as the sole route to budget, and coalition logic read as a hypothesis language about populations — all have non-trivial content at the same time, and in which a century of external calibration is available to say whether they are worth anything.

1.1 What is being claimed, and what is not

We are not claiming that a society of piece-agents will play strong chess. On the contrary: §13 argues it will play weak chess for a long time, for reasons that are structural rather than incidental, and the note is written so that this is not a surprise. What is claimed is that the society is a better instrument than a conventional engine for a specific question the programme has already asked and answered only in miniature, which is whether the yield of a hypothesis is worth its price; and that chess is the smallest board on which the answer could plausibly differ from the answer noughts-and-crosses gave.

1.2 The spectrum, and where the interesting structure is

The design space is usually presented as a spectrum. At one extreme `f1r3lang` coordinates sixteen sessions with different language models, so a piece-agent is a thin wrapper around a model session. At the other extreme each agent is `f1r3lang` logic throughout. Section 10 argues that the extremes are not really two points on one dial. They are two answers to a single question — where a piece’s policy lives, and whether it is priced — and that the programme already has a construction that lets one have both without incoherence, namely the aperture. The recommendation is not a compromise between the extremes but a relocation:

the model goes outside the ply loop, where it is good at the jobs it is good at, and inside only as a declared aperture in a stratified shard.

1.3 Organisation

Section 2 states what is imported from the existing programme and what is assumed to exist at development time. Section 3 argues for chess as the next worked example. Sections 4–9 are the analysis: two societies, the board, the ply as a join, the coordination problem, coalition logic, and material as metabolism. Section 10 handles the model axis. Section 11 is the implementation spec and is the longest section. Section 12 lists the measurements. Section 13 is the honest accounting of what will go wrong. Section 15 lists open problems. Appendix A is the rho listing and Appendix B the formula catalogue.

2 What is imported, and what is assumed

2.1 From the mortal-scientist programme

The note stands on five constructions that already exist and are not re-derived here.

The cut and the three grades of access. A computation is ontologically isolated; the environment of a computation is other computations; and a learner reaches its environment by perception (reading deposited structure, priced by formula depth), interaction (rendezvous, priced by the communication) or reflection (quotation). The grades are separated in practice by a choice of namespace, not by a modality: the reason a threat marker in the noughts listing lives on its own name family is that were a line to signal a threat on a square’s move channel, the signal would rendezvous with the square’s receipt and the line would have *played the move*. Perception must not be able to act.

The generated logic. One primitive modality $\langle K_j \rangle$ per rewrite rule and choice of redex position, with j ranging over redex positions; structural connectives including separating conjunction; the name predicate $@\varphi$; graded separation over a *described namespace*, $\varphi \mid_k^N \psi$; greatest fixed point ν . There is no hidden-name quantifier and no need of one, because *new* is syntactic sugar: a name is a quoted process, no process contains the name that codes it, so a name fresh in P is manufactured by quoting any term properly containing P . Consequently every family of names used below is *described* by a predicate rather than enumerated, which is what makes graded separation over it available.

The assay. A hypothesis is not consulted and then acted on. It sits in guard position in a *where* clause, so acting *is* testing and confirmation *is* the rendezvous. The complementary guard pair yields three outcomes — confirm, refute, and a third that is empty, in which the budget was spent and nothing was learned. The third is ignorance, not refutation, and it is indistinguishable from the environment being faulty. That identification of the two bottoms is imported wholesale.

Conservation and predation. Nothing mints. The arena routes. On a win the arena hands the winner the loser’s metabolic name, so $\sigma + \kappa = \Theta$ throughout. An exogenous reward would need a referee with a mint and a referee with a mint destroys conservation.

Read versus drive. There are two routes to any predicate. Read it, at perception prices, if the environment has already computed it into legible structure; or drive it, priced by rendezvous. This is a claim about environments, not about learners, and it is statable only because the framework prices the grades separately and both parties have stacks. The noughts-and-crosses measurement found that the three non-modal rungs of its ladder delivered +0.444 of a total +0.481 gain — 92.3% of the yield — for 87.9 of 255.9 metered units, that is 34.3% of the cost.

2.2 From the consensus and quorum line

Coalition logic in Gabbay’s presentation, augmented with the behavioural connective of context-labelled modalities, with the graded reading developed in the quorum note: the coalition modality over a namespace N valued in a resolution semiring V ,

$$\langle E \rangle_N^V \psi = \bigoplus_{O \subseteq N \text{ open}} \bigotimes_{q \in O} \psi(q),$$

where opens are described sub-namespaces via the name predicate. Three results are imported. First, opens are not enumerated but described, so $\varphi \mid_k^N \psi$ applies to coalitions. Second, over $\mathbb{R}_{\geq 0}$ with $\otimes = \times$ a large coalition decays toward zero, so the correct quorum semantics is $\otimes = \min$, a coalition being as strong as its weakest member. Third, over the tropical and Viterbi algebras the attainment proposition holds on the nose, so the coalition modality is *attained* and the reader gets a witness coalition rather than a scalar.

The two-quorum theorem of the swarm note is imported in statement and re-proved in §7 in a form specific to this setting, where it is much stronger.

2.3 From the platform

Cost-accounted rho as it stands on `feature/cost-accounted-rho`. Two facts about the native integration bear on the spec. Metering is per-COMM at the reducer, keyed by the deploy envelope’s signature, and a signed term is *recognised* rather than lowered to a parking gate, so the normalised `Par` has the same COMM count as the unsigned program and there is no double-metering. In-program parking does not occur natively; an unfunded deploy is rejected at the acceptance gate rather than parked mid-program. The surface forms are unchanged: `s :: ()` mints, `{% P %}[s]` gates, `(*)` is joint authority, `::` stacks a budget, `-o` delegates, `# P` signs a section.

The live guard fragment is arithmetic and pattern only. Structural and modal guards are marked `///!` `ideal` in every listing in the repository, meaning they specify a check rather than perform one. Appendix A follows that convention exactly.

2.4 What is assumed to exist at development time

The instruction under which this note is written is that an implementation of coalition logic together with context-labelled modalities will be available. We therefore do not design around its absence, but we do state precisely what is being relied on, because a spec that assumes a component should say what the component must do. These are the requirements; they are collected again as an interface in §11.6.

Requirement 2.1 (Satisfaction, not synthesis). The checker must answer $P \models \varphi$. The consensus note’s term assignment runs the other way, $\vdash \varphi \triangleright P$, proofs as processes; the society needs the model-checking direction, because satisfaction is the guard side and a guard side is what a `where` clause is. Coalition logic here gives a piece a way to *ask* whether a coalition

has formed, not a way to build one. This is the reason the whole synthesis-side gap of the consensus note — its missing proof system — is not on this project’s critical path.

Requirement 2.2 (Three-valued verdicts with a budget bottom). Verdicts are Kleene $\{f < b < t\}$ with b carrying both readings: the participant is faulty or stuck, and the observer could not afford to decide. The caller must not be able to distinguish them, and the spec below depends on that indistinguishability being visible in the measurements.

Requirement 2.3 (Semiring-parametric grading with attainment). The value algebra is a parameter. At minimum $\mathbb{B}$, $\mathbb{R}_{\geq 0}$, tropical $(\min, +)$ and Viterbi $(\max, \times)$ must be available. For the tropical and Viterbi instances the checker must return a *witness*: the coalition attaining the $\oplus$, not merely its value.

Requirement 2.4 (Indexed context-labelled modalities). $\langle K_j \rangle$ must be indexed by redex position, and the quantifiers $\exists j$ and $\forall j$ must range over redex positions of the presented term. An action-labelled logic will not do: every claim in the noughts world was the same action, and in chess every quiet move is the same rewrite rule. The distinctions that matter are positional.

Requirement 2.5 (Settled modality). A derived modality $\langle\langle K_j \rangle\rangle$ meaning “the rewrite at j fires, the board’s deposit computation settles, and then φ holds” must be available, with the settling shown terminating and confluent. In the noughts world this was Lemma 8; §5 restates the obligation for chess, where it is harder because ray scans are recursive.

Requirement 2.6 (Costed queries). Every query returns a metered cost alongside its verdict, charged to the caller’s stack, so that the ladder of §11.4 can be priced and the measurements of §12 can be taken. A checker that answers for free would make the entire experiment vacuous.

Requirement 2.7 (Namespace-indexed opens). The coalition modality must accept its namespace as a name predicate, so that “my two rooks” and “the pieces bearing on f7 ” are expressible without enumeration, and so that graded separation over the coalition namespace is available.

Requirement 2.8 (Caching by verdict). Repeated queries of the same formula against an unchanged deposit must be served from cache at the price of a lookup. Sixteen agents will ask overlapping questions; without caching the read-versus-drive measurement of §12 measures the checker’s implementation rather than the architecture.

3 Why chess, and why now

The programme’s existing worked example is noughts-and-crosses. Every number in it is exact over the 255,168 plays, which is its strength, and it is solved and tiny, which is its weakness. Four things are true of chess and false of noughts, and each of them turns a stipulation of the earlier note into a question with an answer.

Material exists, so eating is non-trivial. In noughts nothing is captured; predation happens only at the arena, between games, when a loser’s metabolic name is handed over. In chess predation happens *during* the game, at the granularity of a single ply, and the prey is a peer agent with its own stack. Conservation has to hold across that, and the accounting becomes interesting rather than ceremonial.

Coalitions are real, so coalition logic has content. Chess strategy is already coalition talk: the rook and bishop together force mate, these three pieces hold the kingside, that knight is doing two jobs. In noughts a coalition is a pair of threats and the vocabulary is overkill. In chess the coalition modality has something to quantify over.

The position is legible, so read-versus-drive is a real choice. The attack map of a chess position is expensive, shared, and recomputable. Whether the world deposits it or each agent drives it is a genuine design fork with a measurable cost difference. In noughts the analogous object is eight lines and the question does not bite.

The science cannot be closed, so the ecology does not die. The noughts note found that χ , the exact characteristic formula, is reached at nine unfoldings, and then found that this is fatal: a χ -monoculture draws every game, the harvest rate goes to zero, and the population starves on a finite wall. A solved science is a dead ecology and imperfection is load-bearing. In chess χ is not reachable at any budget the framework admits, so the ecology cannot die that way. We record this as the reason the ecological half of the programme becomes *observable* here for the first time.

Remark 3.1 (Chess keeps the ecology alive). The noughts result that a solved science is a dead ecology was obtained by exhibiting the solved case. It is a theorem about a closed science and its practical content — that real populations must be imperfect to eat — could not be observed in a world where perfection was affordable. Chess supplies a world where it is not. This does not make the earlier result stronger; it makes it testable in a regime where the alternative hypothesis, that the population converges on a single good policy and stops, is actually available.

Against those four, one thing is true of noughts and false of chess: exactness. Nothing in the chess project will be exact over the game tree. Every measurement in §12 is a sampled estimate against a fixed reference opponent, and the note should say so in its own results section rather than discovering it in review.

4 Two societies, one board

The proposal names two things as agents: the sixteen pieces and the board. It is worth being explicit that these are two *societies*, with different jobs, different budgets, and different logical characters.

Definition 4.1 (The world and the player). The *world* is the society of sixty-four square-agents together with the ray, attack, pin and legality machinery they support, plus the arena and the clock. The *player* is the society of at most sixteen piece-agents of one colour. The world holds no strategic stack and mints nothing; it deposits structure and refuses illegal joins. The player holds the budget and makes the decision.

This is the same split the noughts listing already makes, between `Board`, `Cells`, `Line` and `Scan` on one side and `Choose`, `Move` and `Revise` on the other, and it is the split that makes the question *who is billed* askable. The noughts note observed that putting threats in the term relocates the cost of the scan from the scientist's budget to the world's, and that the interesting question is who pays. In chess that question has a large answer, because the object in dispute is the attack map.

Remark 4.2 (The decomposition is a choice, and there are others). Agent-per-piece is not the only decomposition. Agent-per-square is the world. One could also try agent-per-ray

(the sliding lines as agents, which is close to how the noughts Line contract behaves), or agent-per-plan, where an agent is a coalition and pieces are its resources. Agent-per-piece is chosen because it is the decomposition in which predation is definable: a piece can be captured and a ray cannot. The alternatives are worth naming because agent-per-plan is what a strong human player's introspection reports, and it is the natural second experiment.

5 The board

5.1 Why one name family per square is not enough

The proposal's encoding is

$$\prod_{1 \leq i, j \leq 8} @ (i, j) ! (\cdot)$$

with the payload **0** for unoccupied and $*x$ for the fresh channel minted to interact with a particular piece. The occupancy half of this is right and the manufactured-name half is right — we return to both — but the encoding conflates two roles that the existing listings deliberately separate, and the reason they separate them is stated in the noughts source: perception must not be able to act, and that separation of access grades is enforced by choice of namespace.

In noughts the separation is cheap because perception is short-range: eight lines read three cells each. Chess has genuine long-range perception. A rook on `a1` must know what stands on `a8` without any intention of going there, and a bishop's diagonal passes over squares it will never occupy. If reading occupancy means consuming and re-emitting $@(i, j)!(v)$, then every perception is billed as an interaction, sixteen agents scanning concurrently serialise on every square, and the read-versus-drive distinction — which is one of the two things this project exists to measure — has been designed out of the system before the first line of code.

Definition 5.1 (Three name families per square). Let b be the board's channel and s a square index. The board maintains:

the occupancy cell

$occ(s) = @[*b, "occ", s]$, a resting send carrying either **0** or $*p$ where p is the piece-agent's channel. This is the only family a ply consumes.

the attack family

$atk(s, c, p) = @[*b, "atk", s, c, p]$, standing offers deposited by the settling computation, one per attacker p of colour c bearing on s . Read-only in practice: no move protocol receives on it.

the constraint family

$pin(s)$, $chk(c)$, $ep(s)$, $cast(c, side)$, likewise deposited, carrying the derived facts a legality guard consults.

All three are manufactured names in the sense of the freshness-by-quotation proposition, not `new-bound`, so each family is describable by a name predicate and graded separation over it is available.

Proposition 5.2 (Perception must not act). *If attack markers are deposited on the occupancy family, then a scan that signals “ p bears on s ” can rendezvous with a move protocol receiving on $occ(s)$, and the scan performs a ply. Separating the families makes that failure unrepresentable rather than merely unlikely.*

Proof. Immediate, and it is the chess instance of the argument written into the noughts source for threat offers. The content is that the separation is structural: with distinct families there is no term in which a scan and a move protocol can rendezvous, so the property does not depend on any discipline the scan follows. $\square$

5.2 Who is billed for the attack map

The attack map is the expensive object in chess and it is shared by every agent in the player. There are two architectures.

World-billed.

After each ply the board runs a settling computation that walks every ray and deposits *atk* markers. The player's agents then read markers at perception prices. One computation, sixteen readers.

Player-billed.

The board deposits only occupancy; each agent that needs to know whether a square is attacked drives the query itself, paying a rendezvous per square walked. Sixteen computations of one fact.

The framework's own thesis says the first should win, and by a factor near the population size, because a predicate the environment has computed into legible structure is read at formula-depth prices rather than driven at rendezvous prices. But the framework does not say this is free: the world's settling computation is metered too, and the noughts note's remark on the point is precisely that the cost is *relocated*, not removed, and that the interesting question the framework can pose — because both parties have stacks — is who is billed. We therefore specify both and measure the difference (measurement M7, §12). This is the cheapest genuinely informative experiment in the whole project.

Remark 5.3 (Settling in chess is harder than in noughts). The noughts settling lemma had an easy proof: each of eight lines fires once, no line's output enables another because offers go to names no line reads, so the computation terminates and is confluent. The chess settling walks rays, and a ray's walk is recursive in the board's occupancy: a slider's ray terminates at the first occupied square. Termination is still clear — each ray step consumes one square of a strictly decreasing remaining ray — and confluence still holds for the same reason, that markers are deposited on a family nothing in the settling reads. But the obligation must be discharged explicitly, because $\langle\langle K_j \rangle\rangle$ depends on it and so does every modal rung of the ladder.

5.3 Legality is a universal, and it belongs to the world

Check, pin, and discovered check are constraints on the legality of *every* piece's move, and they are global: whether the bishop on c4 may move depends on whether doing so exposes the king. No individual piece can establish this locally and no quorum of pieces can establish it either, because an illegal move is not a move and a vote does not make it one.

Principle 5.4 (The legality universal is the world's obligation). *The one unavoidable $\forall$ in chess is placed on the world's budget, not the player's. The world refuses illegal joins, the way the arena in the noughts world holds no stack and only routes. The consequence is that the player's own hypothesis language stays existential and finitely confirmable, which is the regime in which the asymmetry proposition says witnesses are cheap.*

There is a second consequence, and it is the one that makes this satisfying rather than merely convenient. If the world deposits $pin(s)$ markers — computed by walking outward from each king through the first own piece to the first enemy slider — then pin-awareness becomes perception. A piece does not reason about whether it is pinned; it reads whether a marker stands on its square. Likewise check is a marker, not a search. The single most classically “knowledge-heavy” part of chess legality becomes deposited structure, and the read-versus-drive thesis gets its best instance.

Remark 5.5 (What the world does not deposit). The world deposits facts about the *current* position: occupancy, attacks, pins, checks, en-passant and castling rights. It does not deposit facts about *consequences* of moves, because those are the player’s hypotheses and pricing them to the world would smuggle the evaluation function back into the architecture through the floor. The line between the two is exactly the line between the non-modal and the modal rungs of the ladder, and it should be guarded in review: any proposal to have the board deposit “this move loses a piece” is a proposal to write a conventional engine and call it an ecology.

6 A ply is a join

6.1 What the noughts construction got, and why chess does not inherit it

The second draft of the game note was a correction: the context K had never been instantiated, and instantiating it as a *redex* context was what made $\langle K_s \rangle$ a generated primitive rather than a derived abbreviation. The payoff was a bijection. An empty square is a receipt offering to be claimed; a claim is one join consuming the mover’s send and the stale mark; after it fires no receipt remains. So redex positions correspond to empty squares correspond to legal moves, and the branching factor of $\exists s$ is the legal move count.

None of that transfers to chess for free. A chess move touches two squares at minimum, vacating one and occupying another. A capture additionally ends a sibling agent’s computational life. Castling touches four squares and two pieces. En passant has a destination distinct from the square of the captured pawn. Promotion replaces a piece’s identity. If a ply is several rewrites then $\langle K_j \rangle$ indexes something finer than a move, the bijection fails, the modal rungs of the ladder index the wrong thing, and — worse operationally — two plies can interleave in the middle of each other.

6.2 The repair

The repair is available and it is already demonstrated in the platform. The cost-accounting demonstration uses an N -ary atomic join for a four-cell swap that conserves money and goods all-or-nothing in every interleaving, with the guard deciding before any leg fires. A chess ply is that shape exactly.

Definition 6.1 (A ply). A ply is a single N -ary join over the occupancy cells it touches, guarded by a *where* clause consulting the constraint family, which re-emits every consumed cell with its updated payload. Formally, for a move m touching squares $s_1, \dots, s_n$,

$$\text{for } (@v_1 \leftarrow occ(s_1) \ \& \ \dots \ \& \ @v_n \leftarrow occ(s_n) \text{ where } \text{Legal}(m, \vec{v})) \{ \dots \}$$

where the body re-emits $occ(s_1), \dots, occ(s_n)$ with the post-move payloads, terminates any captured piece-agent, and triggers the settling.

Proposition 6.2 (Move taxonomy is join arity). *Under Definition 6.1 the classification of chess moves by arity and payload transformation is exactly the classical move taxonomy:*

Move kind	Arity	Payload transformation
Quiet move	2	$(*p, \mathbf{0}) \mapsto (\mathbf{0}, *p)$
Capture	2	$(*p, *q) \mapsto (\mathbf{0}, *p), q \text{ terminated}$
En passant	3	$(*p, \mathbf{0}, *q) \mapsto (\mathbf{0}, *p, \mathbf{0}), q \text{ terminated}$
Castling	4	$(*k, \mathbf{0}, \mathbf{0}, *r) \mapsto (\mathbf{0}, *k, *r, \mathbf{0})$
Promotion	2	$(*p, \cdot) \mapsto (\mathbf{0}, *p'), p \text{ retired}, p' \text{ born}$

We state this as a proposition rather than a table because it has a consequence.

Corollary 6.3 ($\langle K_j \rangle$ survives). *With plies as joins, redex positions of the ply rule correspond bijectively to legal moves, so $\langle K_j \rangle$ remains a generated modality with j ranging over moves, $\exists j$ has the legal move count as branching factor, and every modal rung of the ladder indexes what it is supposed to index. The whole ladder apparatus of the game note transfers.*

Remark 6.4 (The alternative, and why not). One could let a ply be several rewrites and recover a composite via the settled modality, as the noughts world does for the board's own settling. The cost is that a ply is then no longer atomic in the term, so two agents' plies can interleave, and the guard that makes a move legal would have to be re-checked against a position that may have changed under it. Chess has a turn structure and it should be represented by atomicity rather than by convention. The join route also costs nothing: the platform has N -ary joins and the demonstration of a conserving four-cell swap already exists.

Remark 6.5 (Promotion is birth and it is not free). Promotion is the only move that creates an agent. Under conservation it cannot create a *budget*: the promoted agent's stack must come from somewhere, and the natural source is the pawn's own stack, transferred whole. That makes promotion a change of policy and capability at constant budget, which is the right reading, and it means the classical observation that a promoted queen is worth a queen is not automatic here — it is a prediction, since the promoted agent inherits a pawn's accumulated stack rather than a queen's.

7 The coordination problem

7.1 The shape of the problem

Sixteen agents each want to move; exactly one move commits. This is not a technical inconvenience, it is the whole architectural content of the proposal, and the programme already contains the analysis of it.

Observation 7.1 (The K/J split). A graded clause is a map from candidates to values: a *weighting* of the fibre, not a *section* of it. In the selection-monad reading, a clause is a quantifier $(X \rightarrow R) \rightarrow R$ and a resolver is a selection function $(X \rightarrow R) \rightarrow X$. The swarm exhibits the split as different behaviours performed by different bees: dance evaluation is the clause, piping and takeoff are the section, and no bee does both.

The piece society must exhibit the same split. A piece *evaluates* candidate moves; something else *commits*. A design in which each piece both scores and plays is a design with sixteen resolvers and no game.

7.2 Quorum, not consensus, and the argument is not a preference

The obvious architecture is a council: all sixteen pieces report scores, the scores are combined, the argmax commits. That architecture has a chairman, and a chairman is a global

evaluator wearing a hat. The programme's altitude argument rules it out, and it does so by derivation rather than by taste.

Proposition 7.2 (The mechanism is forced). *Consensus at the council is a global $\forall$ over all sixteen advocates with no presented witness: finitely refutable, not finitely confirmable. Quorum at a candidate move is a local $\oplus \otimes$ over one open. No participant can evaluate the global universal within a bounded budget, so a society of budget-bounded participants must use the local formula.*

This is the quorum note's derivation of Seeley's empirical finding, restated for a designed rather than an evolved population. In the swarm it explains why quorum is sensed at the site rather than at the cluster. Here it says: pieces advocate for candidate *moves*, and a candidate that accumulates enough advocacy commits. Nobody counts the whole distribution.

7.3 The advocacy namespace, and why a candidate move is a name

Definition 7.3 (Advocacy). A candidate move m is a term; by reflection $@m$ is a name. The advocacy channel for m is $adv(m) = @[*pl, "adv", @m]$ where pl is the player's channel. A piece advocates for m by sending its advocacy token on $adv(m)$. An *open* in the coalition sense is the set of pieces that have advocated for a given m during the current round; it is described by a name predicate, not enumerated.

This is the herd note's corollary landing exactly: opens are described namespaces via $@\varphi$. It is also, we think, the most pleasing single consequence of working in a reflective calculus. In the swarm an open is a set of bees and the modeller has to say so. Here the site a piece advocates for *is* a name, manufactured by quoting the move, and the advocacy set is a sub-namespace of the player's namespace. Nothing has to be introduced to make coalition logic apply; the calculus already provides the points.

7.4 One token per piece per round: the society is a polity

Definition 7.4 (The franchise). At the start of each advocacy round the player issues to each surviving piece exactly one *linear* advocacy token, of signature V , into a located stack whose channel is manufactured from the piece's identity. A piece spends its token by advocating, or by inhibiting, or not at all. Tokens do not carry over between rounds.

This is the polity listing's pre-election phase transposed without change: a persistent claim on the roll metered to one linear token per epoch, a source keyed to identity and not a mint. The reason to adopt it is not tidiness. It is that it makes the following theorem an arithmetic fact about sends rather than a statistical regularity.

7.5 Quorum is a join arity

Definition 7.5 (The commit gate). The commit gate for candidate m at threshold k is the k -ary join

$$\text{for } (@a_1 \leftarrow adv(m) \ \& \ \dots \ \& \ @a_k \leftarrow adv(m)) \ \{ \text{Commit}(m) \}$$

which fires exactly when k distinct advocacy sends stand on $adv(m)$, consuming all of them.

Theorem 7.6 (Two quorums need $2k$ tokens). *Let S be the number of surviving pieces, each holding exactly one linear advocacy token per round, and let k be the commit threshold. Then two distinct candidate moves cannot both fire their commit gates in the same round unless $S \geq 2k$. In particular with $S = 16$ and $k \geq 9$ a split is impossible.*

Proof. Each commit gate consumes k sends on its own channel. Tokens are linear, so a token spent on $adv(m)$ is not available on $adv(m')$ for $m' \neq m$, and each piece holds one. Two firings therefore consume $2k$ distinct tokens drawn from a pool of S . $\square$

Remark 7.7 (What this theorem is worth). In the swarm the corresponding statement is a structural impossibility result that a statistical model cannot state, and it was confirmed by simulation: splits identically zero at $S = 12, 16, 20, 24, 28, 29$ and first appearing at $S = 30 = 2k$ with $k = 15$. Here it needs no simulation, because linearity of the token and arity of the join do the counting. That is the difference between importing a result and re-deriving it in a calculus that already had the right primitives. It also means the designer has a free safety property: choose $k \geq \lceil S/2 \rceil$ and the society *cannot* commit two moves, for any policies whatsoever, with no cross-inhibition machinery at all.

7.6 The other failure mode, and what cross-inhibition is for

Safety comes free; liveness does not. With $k \geq 9$ and advocacy split eight against eight, no gate fires and the round deadlocks. The swarm literature's division of labour applies exactly: below $2k$ cross-inhibition prevents *deadlock*; above $2k$ it prevents *split*. Since the design fixes $k \geq S/2$ and buys no-split structurally, cross-inhibition in this setting has only the deadlock job, and it competes with a cheaper mechanism.

Definition 7.8 (Rounds, and the deposited distribution). A turn consists of up to R advocacy rounds. At the end of a round in which no gate fired, unconsumed advocacy sends remain standing on their $adv(m)$ channels and are *readable*. Fresh tokens are issued. A piece may therefore re-advocate having perceived, at perception prices, where the rest of the society stood.

This is the bees reading dance strength at the cluster: cheap perceptual price for the reversible decision (which candidate to back), expensive interaction price for the irreversible one (the commit). We take that as the default and treat cross-inhibition as a labelled variant, for two reasons. It is one more mechanism, and the swarm measurement recorded an unexpected cost — with cross inhibition on, the swarm chose the better site slightly *less* often, because inhibition suppresses correct advocacy too.

Definition 7.9 (The resolver of last resort). If R rounds elapse with no commit, the clock commits the candidate carrying the most standing advocacy, ties broken by a deterministic order on $@m$.

This is the one centralised element in the design and the note should not hide it. It is held by the clock, which is world and not player; it has no evaluation function and no preferences; and its firing rate is a measurement (M4, §12), because a society that reaches last resort often is a society whose quorum parameters are wrong.

7.7 The population shrinks, and it crosses the boundary during play

Proposition 7.10 (The endgame crossing). S is monotonically non-increasing over a game, from 16 toward as few as one piece besides the king. If k is held fixed as an absolute number, then $S < k$ eventually holds and no gate can ever fire; if k is held fixed as a fraction αS with $\alpha > 1/2$, splits remain impossible throughout but the absolute number of advocates required falls. Either way the society crosses a regime boundary at a computable ply.

Remark 7.11 (A prediction the bees could not supply). The swarm's S is fixed within an episode, so the two-quorum theorem there partitions *episodes*. Chess partitions the *inside* of

a single episode: the same society passes through the split-possible regime and into the split-impossible one as material comes off. The framework therefore predicts that the character of the decision procedure must change at a computable point, with competing plans available in the middlegame and structurally unavailable in the endgame. That this matches ordinary chess intuition — middlegames are about competing plans, endgames about executing one — is not evidence, but it is the kind of coincidence that makes a measurement worth taking rather than a post-hoc story worth telling. M5 in §12 is that measurement, and it is the one we would report first.

8 Coalition logic as the society’s strategic vocabulary

8.1 The vocabulary is already chess’s own

Chess commentary is coalition talk. The rook and bishop together force mate. These three pieces hold the kingside. That knight is overloaded, doing two jobs at once. That bishop is bad, belonging to no coalition. Coalition logic’s $\langle E \rangle \varphi$, “coalition E can force φ ”, is not a formalism imported and then applied; it is the native register of the domain.

This matters for the project because it means the society’s hypotheses can be *about* coalitions without anyone having designed a coalition-formation mechanism. Pieces advocate; advocacy sets are opens; opens are coalitions; and the logic evaluates formulae over them.

8.2 Which algebra, and why the answer is not $\mathbb{R}_{\geq 0}$

Proposition 8.1 (Large coalitions must not decay). *Over $\mathbb{R}_{\geq 0}$ with $\otimes = \times$ and values in $[0, 1]$, a coalition’s value $\bigotimes_{q \in O} \psi(q)$ is non-increasing in $|O|$ and strictly decreasing whenever any member’s value is below one. Hence larger quorums score weaker, which inverts the intended semantics.*

Definition 8.2 (Quorum algebras). A quorum algebra takes $\otimes = \min$: a coalition is as strong as its weakest member. Over the tropical $(\min, +)$ and Viterbi $(\max, \times)$ algebras the attainment proposition holds on the nose, so $\oplus$ is realised by a definable selection function.

We note, because it is a pleasant piece of evidence rather than an argument, that $\otimes = \min$ is independently the correct chess semantics. A combination fails at its weakest participant. If three pieces bear on a mating square and one of them is pinned, the combination is worth what the pinned piece is worth, which is nothing; the other two do not compensate. Overloading is the same shape from the other side: a piece that belongs to two coalitions caps both at its own availability. The algebra the framework forces on general grounds is the algebra the domain would have chosen.

Corollary 8.3 (The player gets a witness). *Over the attained algebras the coalition modality returns not a scalar but the coalition realising it. The society therefore commits to a move together with the coalition that justifies it.*

Remark 8.4 (Explanation is structural, not decorative). Corollary 8.3 is the closest thing this project has to a by-product worth having on its own. An explanation of the form “this move, because these pieces, at this strength” is not generated after the fact by a separate narration module reading the engine’s numbers. It is the witness the selection function returned. Whatever else is true of the resulting player, its explanations cannot be confabulated, because there is no separate faculty available to confabulate with.

8.3 The alternating rung, and the question this project exists to answer

Coalition logic’s “can force” quantifies over the opponent’s replies. It is therefore an alternating formula, a box under a diamond, and the game note’s measurement of the corresponding rung was unkind: rung 4, the one genuinely alternating rung of the noughts ladder, bought +0.005 for 111 metered units and was the worst purchase on the ladder, while the three non-modal rungs delivered 92.3% of the yield for 34.3% of the cost.

We do not think this settles anything, and the reason is the reason for the whole project.

Observation 8.5 (The yield result is a measurement on one game). Noughts-and-crosses has 5478 reachable positions, no material, no long-range interaction, and a branching factor that starts at nine and falls. Its non-modal predicates — a completing threat, a fork, centrality — are close to sufficient because the game is close to trivial. Whether the finding generalises is not a question the noughts data can answer, because the finding could be a fact about games or a fact about the smallness of that game.

Principle 8.6 (Chess is the discriminating instance). *The measurement that would most change the programme’s view of itself is the ladder measurement (M1, M2) repeated on chess. If the non-modal rungs again deliver most of the yield for a third of the cost, the read-versus-drive thesis acquires its second and much harder instance and becomes a claim worth defending in general. If the modal and coalition rungs dominate in chess, then the noughts result is a result about noughts, and the programme has learned something it could not have learned from a game it could solve.*

This is the strongest strategic argument for building the thing, and it should appear in the note’s own abstract, which it does.

8.4 Graded separation and what a fork is in chess

The noughts fork was $\text{Threat}(p) \mid_0^N \text{Threat}(p)$: two threats sharing no square, graded at separation zero over the square-indexed namespace. Requiring it is what forced the board encoding, since two threats with a common gap must be excluded and therefore squares rather than lines must be the parands.

The chess transposition is direct and the namespace is the attack family.

Definition 8.7 (Fork, chess instance). Let N_{atk} be the namespace described by the name predicate

$$@[*b, "atk", \cdot, c, \cdot],$$

and write $\text{Bears}(p, s)$ for the structural predicate asserting a marker deposited by p on s . Then

$$\text{Fork}(p) := (\exists s. \text{Bears}(p, s) \wedge \text{Valuable}(s)) \mid_0^{N_{atk}} (\exists s'. \text{Bears}(p, s') \wedge \text{Valuable}(s')).$$

The same argument that forced squares as parands in noughts forces *squares* here too, not pieces: a knight bearing twice on one square is not a fork, and separation over a piece-indexed namespace would not exclude it.

8.5 The coalition rung

Definition 8.8 (The coalition rung). For a target square s and the player’s piece namespace N_{pl} ,

$$\text{Bearing}(s) := \langle E \rangle_{N_{pl}}^V \lambda p. \text{Bears}(p, s) \otimes \text{Available}(p),$$

where $\text{Available}(p)$ is zero if p is pinned or already committed elsewhere. Over a quorum algebra this evaluates to the strength of the strongest available coalition bearing on s , attained at a witness.

Available is where pins enter, and it is worth observing that it enters as a *factor* in the $\otimes$ rather than as a side condition. A pinned piece contributes $\perp$ to any coalition containing it, and under $\otimes = \min$ that zeroes the coalition. This is the correct behaviour and it was not designed in; it follows from the choice of algebra forced in §8.

Remark 8.9 (Brittleness, and the two dials). The herd note recorded a negative result that applies here without change: $\min$ over a large coalition is zeroed by one dissenter, so a crisp rung and a graded rung can return identical rows, and grading pays only if coalition *size* is relaxed as well. The chess instance is that Bearing over a fixed large coalition is brittle, and the design must therefore relax over subsets — which it does, since $\oplus$ ranges over opens, and the witness returned is the best available subset rather than the whole. We flag it because it was a surprise in the earlier work and would be a surprise again.

9 Material as metabolism

The noughts arena routes and never mints: on a win the arena hands the winner the loser’s metabolic name. In chess, capture is predation at ply granularity, and nothing needs reinterpretation for it to be so.

Definition 9.1 (Capture is harvest). When a ply’s join terminates a piece-agent q , the capturing agent p receives q ’s metabolic name; q ’s remaining stack is thereby routed to p . No tokens are minted and none destroyed, so $\sigma + \kappa = \Theta$ holds across the ply.

Proposition 9.2 (Material advantage is budget advantage). *Under Definition 9.1, a side with more surviving material has, all else equal, both more advocates and more aggregate stack, hence can afford more perception per turn and more rounds of advocacy. The conventional evaluation term “material” becomes a statement about the society’s computational capacity rather than a number added to a score.*

This is the most interesting thing in the design and also the least certain.

Conjecture 9.3 (Emergent piece values). *Let $\pi(t)$ be the average lifetime harvest — tokens acquired by capture, plus tokens the agent’s survival makes available to the society — of an agent of piece type t over a population of games. Then the ratios $\pi(\text{pawn}) : \pi(\text{knight}) : \pi(\text{bishop}) : \pi(\text{rook}) : \pi(\text{queen})$ approximate the classical $1 : 3 : 3 : 5 : 9$.*

We state it as a conjecture and not as a design goal, and the difference matters. Nothing in the architecture hands a piece its value. A queen is an agent whose move-generation contract reaches further, which means it can advocate for more candidates, bear on more squares, and belong to more coalitions; and under $\otimes = \min$ belonging to more coalitions is worth something only if the agent is available. If the classical ratios fall out, the framework has derived one of the few quantitative facts about chess that has no derivation. If they do not, the discrepancy is data about which of the framework’s pricing choices is wrong, and it is much more informative than agreement would be.

Remark 9.4 (Two more in the same currency). Offered as shapes rather than claims, because they are the sort of thing the design makes sayable and it would be a waste not to say them. A *pinned* piece is one whose budget is committed but unspendable, which is why Available zeroes it. *Zugzwang* is being obliged to spend a token when every available expenditure is loss-making, which in this setting is not a metaphor: the franchise is issued per round and a piece that must advocate for something is a piece that cannot decline the market.

10 Where the policy lives

10.1 The extremes are not a dial

A piece's policy is the thing that maps the position to an advocacy. The question is where that map lives and whether it is priced. The sixteen-model-sessions extreme and the all-f1r3lang extreme are two answers, and they are not near-neighbours on a continuum.

A model session is opaque. Its choice function is not a clause, so it cannot sit in guard position and the assay construction does not apply. It is not priced, so cost accounting measures API calls rather than computation, and every measurement in §12 loses its denominator. It is not observable, so the coalition formulae become commentary on a decision they did not make, and Remark 8.4 is false — a separate narrating faculty is exactly what a model wrapper reintroduces. At that extreme every structural result in this note evaporates and what remains is a multi-agent chess demonstration, of which there are many. Sixteen sessions per advocacy round is also slow enough that the round structure of §7.8 becomes unusable.

10.2 The aperture, which is the programme's own construction

None of that makes model-backed pieces incoherent, and the reason is that the choice-formulae note already worked out how strength enters a shard from outside. Its Route E is the *aperture*: a channel fed by the environment rather than by the term. Two things are said about apertures there. They are where choice of full strength can enter, and they are syntactically conspicuous, so a static analysis of aperture channels bounds a whole shard. Under the stratified *where* discipline, subject reduction holds and full-strength choice occurs exactly at declared apertures while the interior's choice type stays invariant.

A model session is an aperture, precisely and not by analogy. So a society with model-backed pieces is coherent in a stratified shard, and building it would *demonstrate* the stratification result on an artefact people care about rather than on a counterexample two lines long.

Principle 10.1 (Relocate, do not compromise). *The recommendation is not a midpoint between the extremes. It is that the model goes outside the ply loop by default, and inside only as a declared aperture in a labelled variant.*

Outside the ply loop there are two jobs a model is actually good at, and both are already roles in the existing design. The first is authoring and mutating piece policies between games, which is the *Revise* and *Crossover* and *Beget* role of the noughts listing, and which lands squarely on the reproduction note's crossover over quoted code — a model is a mutation operator on programs, which is a thing models do well. The second is reading the witness coalition out loud, which is post-hoc narration of a decision that was genuinely made elsewhere and is therefore honest.

Remark 10.2 (What the aperture variant would measure). Worth building second, not first, and worth building for a specific reason: it is the only configuration in which one can measure what a model's judgement is worth *in the framework's own currency*. Run the same society with j of sixteen pieces aperture-backed and $16 - j$ f1r3lang-backed, and the yield-per-unit metric prices the model's contribution against the ladder rungs. Nothing else in the current AI-evaluation landscape prices a model's judgement against a formally specified alternative on the same budget.

11 The implementation spec

This section is written to be handed to a developer. It fixes names, protocols, prices, interfaces, delivery phases and acceptance criteria. Where a decision is genuinely open it is marked OPEN rather than resolved silently.

11.1 Module inventory

Module	Society	Responsibility
Board	world	occupancy cells; the ply join; refusal of illegal joins
Settle	world	ray walks; deposit of <i>atk</i> , <i>pin</i> , <i>chk</i> , <i>ep</i> , <i>cast</i>
Arena	world	game lifecycle; harvest routing; adjudication
Clock	world	turn and round structure; resolver of last resort
Registry	world	the roll: which piece channels are live this turn
Franchise	player	issue of one linear advocacy token per live piece per round
Piece	player	per-agent policy: generate candidates, evaluate, advocate
Gate	player	the k -ary commit join, one instance per live candidate
Ladder	player	the formula catalogue; rung held per agent
Revise	player	per-agent rung revision from assay outcomes
Metab	both	Live: reclaim, debit, reissue; death on insufficiency
Check	world	the coalition-logic and modal checker service (§11.6)

Check is placed in the world deliberately. It is an instrument the player *holds*, and its queries are charged to the caller, but it is not part of any agent's term; otherwise sixteen agents would carry sixteen copies of the checker and the caching requirement could not be met.

11.2 Namespace discipline

Every name below is manufactured by quotation, not *new-bound*, so every family is described by a predicate. *b* is the board channel, *pl* the player channel, *g* the game index.

Family	Name	Payload
board	@[*arena, g]	—
occupancy	@[*b, "occ", s]	0 or *p
attack	@[*b, "atk", s, c, p]	attacker descriptor
pin	@[*b, "pin", s]	pinning ray descriptor
check	@[*b, "chk", c]	checking piece set
en passant	@[*b, "ep", s]	capturable pawn channel
castling	@[*b, "cast", c, side]	right present or absent
piece	@[*b, "p", id]	—
piece stack	@[*b, "m", id]	σ , the metabolic stack
advocacy	@[*pl, "adv", @m]	advocating piece id
inhibition	@[*pl, "inh", @m]	inhibiting piece id
franchise	@[*pl, "V", id, r]	the round's linear token
commit	@[*pl, "commit"]	the chosen @m

Requirement 11.1 (No family is receivable by the wrong society). No contract of the player receives on `occ` except through the `ply join` of `Board`; no contract of the world receives on `adv`, `inh` or `V`. This is Proposition 5.2 discharged as a code review checklist item, and it is the single most important invariant in the system.

11.3 The turn protocol

A turn proceeds as follows. Steps marked $\star$ are metered to the player; steps unmarked are metered to the world.

1. **Settle.** `Settle` walks every ray from every occupied square and deposits the `atk` family; then walks outward from each king and deposits `pin` and `chk`. Terminating and confluent by Remark 5.3. On completion it signals the clock.
2. **Roll.** `Registry` publishes the set of live piece channels for the moving colour.
3. **Franchise.** `Franchise` issues one linear `V` token per live piece into that piece's located stack. Round counter $r := 1$.
- $\star$ **Generate.** Each `Piece` enumerates its own pseudo-legal moves from its own square by its own move contract. No piece enumerates another's. The union over pieces is the candidate set; nobody computes the union.
- $\star$ **Evaluate.** Each `Piece` evaluates its candidates against the rung it currently holds, by querying `Check`. The query is an assay: the formula sits in guard position, so evaluating is testing.
- $\star$ **Advocate.** Each `Piece` spends its `V` token on $adv(m)$ for its best candidate m , or declines. OPEN: whether a piece may advocate for a move it does not itself make — see §11.9.
4. **Gate.** A `Gate` instance stands on each live candidate. The first to accumulate k advocacy sends fires and writes @m to `commit`. By Theorem 7.6 at most one fires when $k \geq \lceil S/2 \rceil$.
5. **Round or resolve.** If no gate fired and $r < R$, then $r := r + 1$ and control returns to step 3, with unconsumed advocacy left standing and readable. If $r = R$, the clock applies Definition 7.9.

6. **Ply.** Board performs the N -ary join of Definition 6.1, guarded on the constraint family. A refused guard leaves the message in the tuple space, which is the platform's own semantics for a failed `where`, and the clock retries with the next-best candidate.
7. **Harvest.** If the join terminated an agent, its metabolic name is routed per Definition 9.1.
- ★ **Revise.** Each piece updates its rung from its own assay outcomes by `Revise`. Refutation rate is the temperature, as in the noughts world; it is an observable the assays already emit rather than a schedule imposed from outside.
8. **Adjudicate.** Arena tests for mate, stalemate, and the draw conditions, and either ends the game or passes the turn.

Remark 11.2 (Where the cost actually goes). Steps 5 and 11 are the player's whole expenditure and step 1 is the world's. That is the entire content of measurement M7. Note that step 4 is charged to the player but is cheap and irreducible — a piece must know its own moves — and step 7 is charged to the world because a join firing is a COMM, which under the native integration is metered at the reducer on the deploy envelope.

11.4 The ladder

Each piece holds a rung. Rungs are cumulative and evaluated highest-priority first, as in the noughts listing. `Emp`, `Bears`, `Valuable`, `Available`, `Central` and `Develops` are structural predicates over the deposited families; $\langle\langle K_m \rangle\rangle$ is the settled modality at the ply m .

Rung	Character	Formula added	κ
0	—	$\top$ (any legal move)	0
1	non-modal	$\exists m. \text{Capture}(m) \wedge \neg \text{Defended}(\text{to}(m))$	2
2	non-modal	$\vee \neg \text{Hangs}(m)$: the destination is not attacked by a cheaper piece	2
3	spatial	$\vee \langle\langle K_m \rangle\rangle \text{Fork}(me)$, Definition 8.7	4
4	alternating	$\vee \forall m'. \langle\langle K_m \rangle\rangle [K_{m'}] \neg \text{Fork}(them)$	8
5	spatial, depth 0	$\vee \text{Central}(\text{to}(m)) \vee \text{Develops}(m)$	1
6	coalition	$\vee \langle\langle K_m \rangle\rangle \text{Bearing}(s) \geq v$, Definition 8.8	16
χ	fixed point	$\nu X. \forall m. [K_m](\neg \text{Lost} \wedge \exists m'. \langle\langle K_{m'} \rangle\rangle X)$	32

The prices are the noughts schedule $\kappa(d) = 2^d$ per candidate examined, charged by the depth of the formula the rung installs, with rung 6 priced one step above rung 4 because the coalition modality quantifies over opens and is a width engine. Rung 5 is priced at 1 because it is name-level depth zero, and that pricing is what made it the best marginal purchase in the noughts measurement; whether it repeats here is M2.

Remark 11.3 (χ is unreachable, and that is the point). In noughts χ was exact at nine unfoldings because the board loses a square every step. Chess has no such bound. χ is therefore listed for completeness and as a limit object; no agent will hold it, and by Remark 3.1 that is exactly the condition under which the ecology stays alive.

11.5 The cost schedule

Operation	Grade	Price
read one deposited marker	perception	1
evaluate a rung- d formula, one candidate	perception	$\kappa(d) = 2^d$
cached query, unchanged deposit	perception	1
enumerate own pseudo-legal moves	perception	1 per square walked
spend an advocacy token	interaction	2
fire a commit gate	interaction	$2k$
perform a ply join of arity n	interaction	$4n$
one ray step in <i>Settle</i>	world	1
harvest	interaction	0 (routing, not work)

Every price here is a CHOICE, not a derivation, and the note should say so in the same breath as reporting any number computed with it. The noughts work varied its price scale over three settings and found the shape of its results robust and the boundaries not; the same sensitivity analysis is required here and is part of the acceptance criteria in §11.8.

11.6 The checker interface

The service assumed in §2.4 is used through one contract.

```
// verdict in {"t","b","f"}; witness is Nil, a redex position, or a
// coalition (a list of piece ids); cost is charged to the caller's stack.
contract Check(@formula, @posHandle, @algebra, payer, ret) = {
  // ret!((verdict, value, witness, cost))
}
```

formula

a quoted formula of the generated logic extended with $\langle E \rangle_N^V$, with namespaces given as name predicates.

posHandle

a handle on the current deposit, not a copy of the position. The checker reads the deposited families; it never drives the board. This is Requirement 2.1 enforced by the interface shape.

algebra

one of "B", "Rnn", "trop", "vit". Witnesses are returned for the last two.

payer

the caller's metabolic channel. A caller that cannot fund the query receives verdict "b", which is the budget bottom and is deliberately indistinguishable from a stuck environment.

Requirement 11.4 (The checker may not move). The checker's implementation must receive only on the *atk*, *pin*, *chk*, *ep* and *cast* families and must never receive on *occ*. Violating this makes the pride move the zebra by looking at it, in the phrase the herd note used for the same hazard, and it is the reason the noughts listing insisted on *peek* rather than a consuming receipt in its own scan.

11.7 Reproduction and the offline loop

Between games the population evolves. This is the noughts world's `Pick / Crossover / Beget` apparatus and it transfers with one change of emphasis. There are two loci, the rung and the policy, and the noughts measurement showed that an individual can reach the first by deepening and cannot reach the second at all — the opening changed only across generations, by crossover, because it is a move of distance 2^{-1} in the ultrametric and an individual's incremental revisions are confined to their initial ball.

The chess analogue of “the opening” is the piece's *advocacy style*: how readily it advocates, what it treats as valuable, whether it defers to standing advocacy. That is the locus crossover must reach, and it is where a model-backed mutation operator earns its place (§10).

11.8 Delivery phases

P0 — the world alone.

`Board, Settle, Registry, Clock, Arena`. Acceptance: legal move generation validated against a standard perft suite to depth 5 from the initial position and from a set of tactical positions; the settling shown terminating and confluent by test; every ply a single join.

P1 — one agent, no society.

A single `Piece` contract driving all pieces in turn, rung 0–2, no advocacy. Acceptance: plays legal complete games against a fixed reference opponent; metering totals reconcile against $\sigma + \kappa = \Theta$.

P2 — the society, arithmetic guards.

Sixteen agents, franchise, commit gates, rounds, last resort. Rungs 0–2 and 5 only, since those are the non-modal rungs and are expressible in the live guard fragment. Acceptance: Theorem 7.6 confirmed empirically at k below and above $S/2$; last-resort firing rate reported.

P3 — the checker.

`Check` integrated; rungs 3, 4, 6 enabled. Acceptance: M1 and M2 produced.

P4 — the economy.

Piece stacks, harvest, metabolic death, the offline reproduction loop. Acceptance: M6 produced; conservation audited per ply.

P5 — the aperture variant.

j of sixteen pieces backed by a model session through a declared aperture, in a stratified shard. Acceptance: M8 produced; the static analysis of aperture channels demonstrated.

P0 through P2 are buildable today on `feature/cost-accounted-rho`. P3 depends on the assumed checker. A syntactically manipulable `Theory` of positions is *not* on the critical path and should not be put on it: the DDL and module syntax are absent from the branch, which is the same dependency `F1R3Skein` is waiting on, and nothing in P0–P5 requires them.

11.9 Open specification items

S1

May a piece advocate for a move it does not itself make? Yes gives genuine coalitions

in the advocacy structure and lets a defender back an attacker's plan; no keeps the franchise interpretation clean and makes advocacy self-interested. Our inclination is yes, because it is what makes the advocacy set a coalition rather than a poll, but it changes Theorem 7.6 not at all and can be decided late.

S2

Is k absolute or a fraction of S ? Proposition 7.10 says this determines whether the endgame is split-free or gate-dead.

S3

Cross-inhibition in or out of the first build? Recommended out, per §7.

S4

Is the opponent a society too, or a conventional engine? Recommended conventional for P1–P4: a fixed yardstick is needed, and the asymmetry is itself the experiment. Society-versus-society belongs after M1.

S5

Do pieces hold individual stacks from P2, or only from P4? Holding them early makes the metering real but couples two sources of variance.

S6

What is the reference opponent? It must be weak enough that the society is not simply crushed and fixed enough that measurements are comparable across phases.

12 The measurements

Eight measurements. Each states what would be reported and, where it matters, what result would embarrass the framework.

M1 — the yield ladder.

For each rung, win/draw/loss against a fixed reference opponent at fixed strength, and cumulative metered cost, over a fixed number of games. Reported as the noughts table was: win rate, cost, and marginal return per metered unit. Unlike noughts these are sampled estimates with confidence intervals, not exact counts, and the note reporting them must say so in its own results section.

M2 — non-modal share.

The fraction of total yield delivered by rungs 1, 2 and 5 and the fraction of total cost they consume, against the noughts figures of 92.3% and 34.3%. This is the discriminating measurement of Principle 8.6. Either outcome is publishable and they say opposite things.

M3 — the coalition rung.

Rung 6's marginal return, and separately whether its witness is *informative*: does the returned coalition coincide with what a strong player would name? Scored by blind comparison on a set of tactical positions. The herd note found that the modality bought irreversibility rather than accuracy — precision 1.0000 at every sampled time, at low coverage — and if that repeats, the yield metric underprices instruments that decline to answer and the second-currency principle acquires its second instance.

M4 — split and deadlock against k .

Firing rate of the commit gate, split rate, deadlock rate and last-resort rate, over k from

2 to 16 at $S = 16$. Theorem 7.6 predicts split rate identically zero for $k \geq 9$, and the prediction is structural, so a single observed split above that threshold is a bug in the franchise’s linearity and should be treated as such.

M5 — the endgame crossing.

Decision character as S falls through $2k$ within single games: rounds to commit, advocacy entropy, and last-resort rate, bucketed by material remaining. This is the measurement Remark 7.11 exists for and the one we would report first, because it is the only prediction here that the source domain could not have made.

M6 — emergent values.

Lifetime harvest per piece type against the classical 1 : 3 : 3 : 5 : 9, per Conjecture 9.3. Report the raw ratios and the regression, and report disagreement loudly if it occurs.

M7 — who is billed.

Total metered cost per game under the world-billed and player-billed attack-map architectures of §5.2, split by society. The framework predicts the world-billed architecture wins on total and that the player’s share falls by a factor near S . This is the cheapest experiment in the set and it directly tests the read-versus-drive thesis.

M8 — the aperture’s price.

With j of sixteen pieces model-backed, yield per metered unit as a function of j , against the all-f1r3lang baseline. Prices a model’s judgement in the framework’s currency.

Remark 12.1 (The counter-move, if there is time). The herd note’s most striking result was the counter-move: an adversary that introduces seams into the population does not induce *error* in the observer, it denies *liveness* — coverage collapsed from 0.7520 to 0.0115 while precision stayed above 0.9986. The chess analogue is an opponent that plays to reduce the society’s advocacy coherence rather than to win material, and the prediction is that such an opponent should drive the society to last resort rather than to blunders. If it holds, illegibility as denial of liveness acquires an adversarial instance, and the legibility trade — that a population buys privacy with the same currency it buys agreement — becomes something one can demonstrate rather than argue.

13 What will go wrong

Branching, and the honest expectation. Chess has a branching factor near thirty-five and rewards depth. A society that pays a rendezvous per candidate cannot search, and the framework’s own answer — read the structure the environment has deposited, do not drive it — means this player will be a perception player rather than a search player. It will lose to anything with a search until its perception is very good. The evaluation criterion must therefore be yield per metered unit, with playing strength as a secondary sanity check, and this must be stated before the first game rather than discovered in the results. A project of this kind fails publicly in exactly one way, which is by having promised strength.

The society is small. The swarm results are asymptotic in scout population; at $S = 16$ the regimes are coarse and the advocacy distribution is lumpy. This cuts both ways, since it is also what makes Proposition 7.10 reachable, but no fine-grained result about quorum dynamics should be expected from sixteen agents.

The guard fragment. Structural and modal guards are not executed by the live reducer; they are specifications. P2 is buildable because rungs 1, 2 and 5 are expressible arithmetically over deposited markers, which is a deliberate design of the ladder rather than a happy accident, but P3 is gated on the assumed checker and the note should not pretend otherwise.

Latency. Sixteen agents, up to R rounds, a checker query per candidate per agent. Even in the all-f1r3lang configuration a turn is many thousands of COMMs. A demonstration has to be watchable, so the round budget R and the candidate cap per agent are the two dials that must be tuned early, and they are both dials that *change the science*, since capping candidates is capping $\exists$. Report them with every number.

Reflection, a fifth time. The subject-reduction hazard the programme has now met four times is present here too: a received process can carry a formula of higher altitude than the receiving clause declares, and this society passes quoted moves and quoted formulae between agents constantly. The stratified *where* discipline is the known repair and it should be adopted from P2 rather than retrofitted, because retrofitting it means revisiting every advocacy message.

14 What is shown and what is not

Shown, in the sense of proved here from imported premises: Proposition 5.2 (perception must not act), Proposition 6.2 with Corollary 6.3 (the join taxonomy and the survival of $\langle K_j \rangle$), Proposition 8.1 and the consequent choice of quorum algebra, and Theorem 7.6 with its corollary that $k \geq \lceil S/2 \rceil$ buys structural no-split. Theorem 7.6 is the only genuinely new theorem in the note and its proof is three lines, which is the correct length: the work was done by choosing a calculus in which linear tokens and N -ary joins are primitive.

Argued but not proved: that the world-billed architecture wins (§5.2, awaiting M7); that legality belongs to the world (Principle 5.4, which is a design principle with a cost argument behind it, not a theorem); that the model belongs outside the ply loop (§10).

Conjectured: emergent piece values (Conjecture 9.3).

Not addressed at all: opening theory and endgame tablebases, both of which are external knowledge and both of which the society has no principled way to consume; the fifty-move and repetition rules, which are history-dependent and therefore want the history monad rather than the deposit; and time control, which would make the budget a clock and is probably the most interesting thing omitted.

15 Open problems

- Q1** Does the noughts yield result survive? This is M2 and it is the project's reason for existing.
- Q2** Is there a principled k ? The design fixes $k \geq \lceil S/2 \rceil$ for safety, but nothing says which value in that range is best, and the swarm's answer — that k is not a number but a rate threshold, scaling with encounter geometry — suggests the chess analogue is a rate as well.
- Q3** Does agent-per-plan beat agent-per-piece? Remark 4.2. The obstacle is that a plan cannot be captured, so predation would have to be redefined.

- Q4** What is the right treatment of the fifty-move and repetition rules? They are the only genuinely historical facts in chess and the programme has a history monad that nothing in this design uses.
- Q5** Does the coalition modality's irreversibility-not-accuracy signature repeat? M3.
- Q6** Can the society learn its own `Open`? The herd note left this open in general; in chess the analogue is whether the pieces can discover which subsets of themselves are worth treating as coalitions, rather than having the namespace predicate handed to them.
- Q7** Does subject reduction hold for the stratified discipline in the presence of quoted moves passed as advocacy payloads? The fifth appearance of the same suspect.
- Q8** Is there a composition law for societies? Two coalitions that each bear on a square are not obviously composable, and the composing-learners note's disjointness-versus-overlap dial is the obvious place to look.
- Q9** What does time control do? Making the budget a clock rather than a stack changes the metabolism from conservation to flow, and it is not obvious the conservation results survive.
- Q10** Does the society's play have a signature a human can recognise? Not a formal question, but the one a reader will ask, and the witness coalitions of Corollary 8.3 are the only handle on it.

16 Conclusion

The proposal was to make each piece an agent. The analysis above says that this is a well-formed thing to do in cost-accounted ρ , that three of its apparently difficult parts are already solved by constructions in hand, and that one of them is easier here than in the setting it came from.

The board encoding wants three name families rather than one, because perception must not be able to act and because the attack map is the shared expensive object whose billing is the project's cheapest experiment. A ply is an N -ary join and the join arity is the move taxonomy, which is what keeps the generated modality generated. Coordination is quorum sensing rather than consensus, and the argument for that is a derivation from budget rather than a design preference. Legality is the one unavoidable universal and it belongs to the world, which has the pleasant consequence that pin-awareness becomes perception.

The part that is easier here than in the swarm is the two-quorum theorem. In the bees it is a structural impossibility result confirmed by simulation. In ρ it is arithmetic: a quorum is a join arity, one linear franchise token per piece per round is the polity's dereliction, and two k -ary joins over a pool of S linear tokens need $2k \leq S$. A designer who sets $k \geq \lceil S/2 \rceil$ has bought a safety property outright, for any policies whatsoever, and has thereby confined cross-inhibition to the single job of breaking deadlock, where a cheaper mechanism — reading the standing advocacy between rounds — is available.

What the project is for is narrower than what it will look like. It will look like a chess engine and it will not be a good one. It is an instrument for one question: whether the finding that non-modal rungs deliver most of the yield for a third of the cost is a fact about games or a fact about a game small enough to solve. Noughts-and-crosses cannot answer that, because everything about it is exact and nothing about it is hard. Chess can, and the same board that answers it also happens to be a world where the ecology cannot die of its own perfection.

A The listing

Conventions follow the game note. A parameter written `c` is a NAME; a parameter written `@d` is DATA; a channel is passed as `*c`, since `@(*c) = c`. Guards marked `//! ideal` use structural or modal connectives that the current reducer does not evaluate; they are specifications of a check rather than checks. Everything else runs. The listing is the spine only: move generation per piece type, the full settling walk and adjudication are elided, since they are long and unsurprising, and the file `chess.rho` accompanying this note carries them.

```
// =====
// A SOCIETY OF PIECE-AGENTS THAT PLAYS CHESS
// =====
//
// Companion listing to "Sixteen Scientists on One Board". Every construction
// of the note appears here as ordinary code:
//
// Sec. 5 the board -- Board, Cells, Settle, Ray (three families)
// Sec. 5.3 legality -- Legal, Pins (the world's ALL)
// Sec. 6 the ply is a join-- Ply (N-ary, atomic)
// Sec. 7 coordination -- Franchise, Advocate, Gate (quorum = arity)
// Sec. 7.6 rounds -- Clock (+ last resort)
// Sec. 8 coalition rung -- Ladder rung 6 (//! ideal)
// Sec. 9 material -- Harvest, Live (conservation)
//
// CONVENTION. A parameter written 'c' is a NAME (a channel); a parameter
// written '@d' is DATA. To pass a channel one sends '*c', since @(*c) = c.
//
// NOTATION. Guards are written in the generated spatial-behavioural logic
// extended with the coalition modality: out(n)phi, in(n)phi, phi | psi, the
// name predicate @phi, the context-labelled modality <K_m>phi, its settled
// form <<K_m>>phi, the greatest fixed point nu X. phi, graded separation
// phi |_k^N over a DESCRIBED namespace N, and <E>_N^V psi over opens of N.
// There is no hidden-name quantifier and no need of one: 'new' is syntactic
// sugar, since a name is a quoted process and no process contains the name
// that codes it, so a name fresh in P is manufactured by quoting any term
// properly containing P. Every family below is manufactured that way, which
// is what makes it DESCRIBABLE and hence what makes |_k^N and <E>_N^V apply.
// The current reducer accepts only the arithmetic and pattern fragment of
// 'where'; guards using a structural, modal or coalition connective are
// marked '//! ideal' and specify a check rather than perform one.
//
// THE THREE FAMILIES (Sec. 5.1). Square s of board b carries:
// occ(s) = @[*b,"occ",s] occupancy; the ONLY family a ply consumes
// atk(s,c,p) = @[*b,"atk",s,c,p] deposited attack markers; read-only
// pin(s) / chk(c) / ep(s) / cast(c,side) deposited constraints
// Were attack markers deposited on occ, a scan signalling "p bears on s"
// could rendezvous with a move protocol receiving on occ(s) and the scan
// would PLAY A PLY. Perception must not be able to act; the separation of
// access grades one and two is enforced here by a choice of namespace, and
// with distinct families the failure is UNREPRESENTABLE rather than merely
// avoided by discipline.
//
// CONSERVATION. Nothing mints. Franchise issues one LINEAR advocacy token per
// live piece per round and does not carry it over. Harvest MOVES a captured
// agent's stack into the capturer's. Every Live debit moves a token from
// sigma into kappa. So sigma + kappa = Theta throughout.
// =====
```

```
new stdout(`rho:io:stdout`), display, arena, wall, player, clock,
  Live, Wall, Board, Cells, Ray, Settle, Pins, Legal, Ply, Harvest,
```

```

Registry, Franchise, Piece, Candidates, Evaluate, Advocate, Gate,
Ladder, Revise, Check, Turn, Round, Adjudicate, Boot
in {

  for (@line <= display) { stdout!(line) }

  // =====
  // METABOLISM -- reclaim, debit, reissue.
  // Between the receipt and the send the stack is a message in flight: living
  // is racing others to your own stack, and the metabolic channel m is
  // exactly the vulnerability that Harvest exploits when a piece is captured.
  // =====

  | contract Live(m, @burn, step) = {
    for (@sigma <- m) {
      match sigma > burn {
        true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
        false => { display!("DIED", "could not fund a step") }
      }
    }
  }

  // A source, not a prey: rate-limited, non-exclusive, non-adaptive, finite.
  | contract Wall(w, @quantum, @reserve) = {
    for (ret <- w) {
      match reserve >= quantum {
        true => { ret!(quantum) | Wall!(*w, quantum, reserve - quantum) }
        false => { ret!(0) | Wall!(*w, quantum, 0) }
      }
    }
  }

  // =====
  // THE BOARD -- occupancy only. Note what is NOT here: a 'new' for the
  // squares. Each square name is the quote of a term properly containing the
  // board, so by the no-self-code theorem it cannot already occur in the
  // board -- locally fresh, no registry consulted, decentralised by
  // construction. And because the family is generated by a predicate on
  // names, the whole namespace can be DESCRIBED rather than enumerated.
  // =====

  | contract Board(b, @placement) = { Cells!(*b, placement, 0) | Settle!(*b) }

  | contract Cells(b, @placement, @s) = {
    match s < 64 {
      true => { @[*b,"occ",s]!(placement.nth(s))
                | Cells!(*b, placement, s + 1) }
      false => { Nil }
    }
  }

  // =====
  // THE SETTLING -- Sec. 5.2 and Rem. 5.3.
  //
  // A ray walk PEEKS at occupancy ('<-') and never consumes it. That is what
  // makes the scan perception rather than interaction: with a consuming
  // receipt the settling would move the pieces by looking at them, and
  // sixteen agents scanning concurrently would serialise on every square.
  //
  // Termination: each step consumes one square of a strictly decreasing
  // remaining ray. Confluence: markers are deposited on the atk family, which
  // nothing in the settling reads. That discharges the obligation <<K_m>>

```

```

// depends on, which in noughts was Lem. 8 and here is harder because the
// walk is recursive in occupancy.
// =====

| contract Ray(b, @from, @dir, @p, @c, @dist) = {
  match (dist > 0) and Onboard(from, dir) {
    true => {
      new here in {
        here!(Step(from, dir)) |
        for (@s <- here) {
          for (@v <<- @[*b,"occ",s]) { // PEEK, never consume
            @[*b,"atk",s,c,p]!("bears", from)
            | match v {
              Nil => { Ray!(*b, s, dir, p, c, dist - 1) } // ray continues
              _ => { Nil } // blocked here
            }
          }
        }
      }
    }
    false => { Nil }
  }
}

// Every occupied square emits its piece's rays; then the kings emit theirs
// to deposit pins. Rem 5.4: this terminates and is confluent.
| contract Settle(b) = {
  new each in {
    Squares!(*b, *each)
    | for (@s <= each) {
      for (@v <<- @[*b,"occ",s]) {
        match v {
          Nil => { Nil }
          _ => { Directions!(v, s, *b) } // per-type ray emission
        }
      }
    }
    | Pins!(*b)
  }
}

// A pin marker is deposited by walking OUT from a king through the first own
// piece to the first enemy slider on the same ray. Pin-awareness is
// therefore PERCEPTION: a piece does not reason about whether it is pinned,
// it reads whether a marker stands on its square. This is the note's best
// instance of read-versus-drive (Sec. 5.3).
| contract Pins(b) = {
  new k in {
    Kings!(*b, *k)
    | for (@(ksq, c) <= k) {
      new d in {
        Rays8!(*d)
        | for (@dir <= d) { PinWalk!(*b, ksq, dir, c, Nil) }
      }
    }
  }
}

// =====
// A PLY IS A JOIN -- Sec. 6, Def. 6.1 and Prop. 6.2.
//
// One N-ary join over the occupancy cells the move touches, guarded on the

```

```

// constraint family, re-emitting every consumed cell with its post-move
// payload. A failed guard leaves the messages in the tuple space, which is
// the platform's own semantics for a failed 'where', so a refusal is a
// no-op and not a rollback.
//
// ARITY IS THE MOVE TAXONOMY:
// 2 quiet move (*p, Nil) |-> (Nil, *p)
// 2 capture (*p, *q) |-> (Nil, *p), q terminated
// 3 en passant (*p, Nil, *q) |-> (Nil, *p, Nil), q terminated
// 4 castling (*k, Nil, Nil, *r) |-> (Nil, *k, *r, Nil)
// 2 promotion (*p, .) |-> (Nil, *p'), p retired
//
// This is what keeps the redex positions of the ply rule in bijection with
// the legal moves, and hence what keeps <K_m> a GENERATED modality rather
// than a derived abbreviation (Cor. 6.3).
// =====

| contract Ply(b, @m, ret) = {
  match Kind(m) {

    // --- quiet move: arity 2 -----
    "quiet" => {
      for ( @vf <- @[*b, "occ", From(m)] & @vt <- @[*b, "occ", To(m)]
        where Legal(m, vf, vt) ) { /// ideal
        @[*b, "occ", From(m)]! (Nil)
        | @[*b, "occ", To(m)]! (vf)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // --- capture: arity 2, and the occupant is EATEN -----
    // Harvest routes the captured agent's metabolic name to the capturer.
    // The arena mints nothing; predation is legal here for precisely one
    // reason, and it is that somebody took a piece.
    "capture" => {
      for ( @vf <- @[*b, "occ", From(m)] & @vt <- @[*b, "occ", To(m)]
        where Legal(m, vf, vt) ) { /// ideal
        @[*b, "occ", From(m)]! (Nil)
        | @[*b, "occ", To(m)]! (vf)
        | Harvest!(Stack(vt), Stack(vf), *display)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // --- castling: arity 4, all-or-nothing -----
    // The same shape as the four-cell SWAP of the cost-accounting demo:
    // the guard decides BEFORE any leg fires, so no interleaving can
    // leave the rook moved and the king not.
    "castle" => {
      for ( @vk <- @[*b, "occ", KFrom(m)] & @v1 <- @[*b, "occ", KTo(m)]
        & @v2 <- @[*b, "occ", RTo(m)] & @vr <- @[*b, "occ", RFrom(m)]
        where CastleLegal(m, vk, v1, v2, vr) ) { /// ideal
        @[*b, "occ", KFrom(m)]! (Nil) | @[*b, "occ", KTo(m)]! (vk)
        | @[*b, "occ", RFrom(m)]! (Nil) | @[*b, "occ", RTo(m)]! (vr)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // en passant (arity 3) and promotion follow the same pattern.
    _ => { ret!(("refused", m)) }
  }
}

```

```

// Conservation: nothing minted, the loser's stack MOVED into the winner's.
| contract Harvest(mLost, mWon, ret) = {
  for (@a <- mLost & @c <- mWon) { mWon!(a + c) | ret!(("ate", a)) }
}

// =====
// THE FRANCHISE -- Sec. 7.4. One LINEAR advocacy token per live piece per
// round, into a located stack whose channel is manufactured from the
// piece's identity. This is the polity listing's pre-election phase
// verbatim: a persistent claim on the roll metered to one linear token per
// epoch. It is a SOURCE keyed to identity, not a mint.
//
// It is also what makes Thm. 7.6 arithmetic rather than statistical.
// =====

| contract Franchise(@ids, @r) = {
  match ids {
    [] => { Nil }
    [id ...rest] => {
      @[*player, "V", id, r]!(("vote", id, r))
      | Franchise!(rest, r)
    }
  }
}

// =====
// A PIECE -- Sec. 11.3, steps 4-6 of the turn protocol.
//
// A piece enumerates ITS OWN moves from ITS OWN square. Nobody computes the
// union of candidates; the union is what the advocacy namespace IS.
// Evaluation is an ASSAY: the rung sits in guard position, so evaluating is
// testing and confirmation is the rendezvous. By Prop. 17 of the scientist
// note the number of candidates a piece can afford to evaluate is its stack
// divided by the price of one check, so a poor piece does not survey its
// own branching.
// =====

| contract Piece(b, @id, @rung, @r, m) = {
  new cand, best in {
    Candidates!(*b, id, *cand)
    | for (@c <- cand where Sat(c, rung, id)) { /// ideal
      best!(c)
    }
    | for (@c <- best) { Advocate!(id, c, r) }
  }
}

// Spending the token IS the advocacy. A candidate move c is a term, so @c is
// a name; the advocacy channel is manufactured from it. An OPEN in the
// coalition sense is the set of pieces that have advocated for a given move
// this round -- described by a name predicate, never enumerated.
| contract Advocate(@id, @c, @r) = {
  for (@tok <- @[*player, "V", id, r]) { /// LINEAR: spent once
    @[*player, "adv", @c]!(id)
  }
}

// =====
// THE COMMIT GATE -- Sec. 7.5, Def. 7.5.
//
// QUORUM IS A JOIN ARITY. The gate is a k-ary join on one advocacy channel;

```

```

// it fires exactly when k distinct advocacy sends stand there, and it
// CONSUMES them. Shown for k = 3; the general case is generated.
//
// THM 7.6. Two gates on distinct candidates consume 2k tokens drawn from a
// pool of S linear tokens, so two quorums require  $S \geq 2k$ . With  $S = 16$  and
//  $k \geq 9$  a SPLIT IS STRUCTURALLY IMPOSSIBLE, for any policies whatever, and
// no cross-inhibition machinery is needed to secure it. Safety is free;
// liveness is not, and that is what Round is for.
// =====

| contract Gate(@c, @k, ret) = {
  for ( @a1 <- @[*player,"adv",@c]
    & @a2 <- @[*player,"adv",@c]
    & @a3 <- @[*player,"adv",@c] ) {
    @[*player,"commit"]!(@c) | ret!("quorum", @c, [a1,a2,a3]))
  }
}

// =====
// THE ROUND -- Sec. 7.6, Def. 7.8 and Def. 7.9.
//
// Unconsumed advocacy is LEFT STANDING and is readable, so a piece may
// re-advocate having PERCEIVED where the rest of the society stood. That is
// the bees reading dance strength at the cluster: cheap perceptual price for
// the reversible decision, expensive interaction price for the irreversible
// one. Cross-inhibition is deliberately absent -- it is one more mechanism
// and the swarm measurement recorded that it suppresses CORRECT advocacy
// too. Deadlock is broken by re-advocacy and, failing that, by the clock.
// =====

| contract Round(b, @ids, @r, @maxR, ret) = {
  new fired in {
    Franchise!(ids, r)
    | Pieces!(*b, ids, r)
    | Gates!(ids, r, *fired)
    | for (@res <- fired) { ret!(res) }
    | for (@t <- Timeout(r)) { /// ideal
      match r < maxR {
        true => { Round!(*b, ids, r + 1, maxR, *ret) }
        /// Resolver of last resort: the CLOCK, not a piece. It holds no
        /// evaluation function and no preferences; its firing rate is a
        /// measurement (M4), because a society that reaches it often is a
        /// society whose quorum parameters are wrong.
        false => { MostAdvocated!(ids, r, *ret) }
      }
    }
  }
}

// =====
// THE LADDER -- Sec. 11.4. Cumulative, highest priority first. The rung
// index is the modal depth of the conjunct it adds, hence the ultrametric
// stage:  $d(\Phi_r, \Phi_{r+1}) = 2^{-r}$ . Prices are  $\kappa(d) = 2^d$  per
// candidate examined, charged by the depth of the formula the rung installs.
//
// Note where the alternation is. Rungs 1, 2 and 5 have no modality at all:
// capturing, not hanging, and centralising are PERCEPTION, reading the
// markers the last settling deposited. The one genuinely alternating rung
// is 4, a box under a diamond, which by Prop. 6 of the scientist note has
// no finite confirming witness. In noughts that rung was also, measurably,
// the worst purchase on the ladder. Whether that repeats in chess is M2,
// and it is the whole reason for the project.

```

```

// =====

//! ideal
// Emp(s) := out(occ(s))@Nil
// Bears(p,s) := exists c. out(atk(s,c,p)) TOP
// Fork(p) := Threat(p) |_0^{N_atk} Threat(p)
// Bearing(s) := <E>_{N_pl}^V lambda p. Bears(p,s) (x) Available(p)
// <<K_m>>phi := the ply join at m fires, the board settles, phi holds
| contract Ladder(@r, ret) = {
  match r {
    0 => { ret!( @{ TOP } ) }
    1 => { ret!( @{ exists m. Capture(m) and not Defended(to(m)) } ) }
    2 => { ret!( @{ exists m. Capture(m) and not Defended(to(m))
                  or not Hangs(m) } ) }
    3 => { ret!( @{ exists m. Capture(m) and not Defended(to(m))
                  or not Hangs(m)
                  or <<K m>> Fork(me) } ) }
    4 => { ret!( @{ exists m. Capture(m) and not Defended(to(m))
                  or not Hangs(m)
                  or <<K m>> Fork(me)
                  or forall m2. <<K m>>[K m2] not Fork(them) } ) }
    5 => { ret!( @{ exists m. Capture(m) and not Defended(to(m))
                  or not Hangs(m)
                  or <<K m>> Fork(me)
                  or forall m2. <<K m>>[K m2] not Fork(them)
                  or Central(to(m)) or Develops(m) } ) }
    // The coalition rung. Over a quorum algebra (x) = min this is attained,
    // so Check returns the WITNESS coalition and not merely its value: the
    // society commits to a move TOGETHER WITH the pieces that justify it.
    6 => { ret!( @{ exists m. exists s. <<K m>> (Bearing(s) >= v) } ) }
    // chi is listed for completeness. Unlike noughts it is NOT exact at any
    // finite unfolding, and by Rem. 3.1 that is exactly the condition under
    // which the ecology does not die of its own perfection.
    7 => { ret!( @{ nu X. forall m. [K m] (not Lost
                  and exists m2. <<K m2>> X) } ) }
  }
}

// Revision walks the lattice. The move has the two parameters the metric
// grades: a LOCUS and a DIRECTION. Refutation rate is the temperature, and it
// is an observable the assays already emit, not a schedule imposed from
// outside. NB no case touches the piece's ADVOCACY STYLE: that is the
// stage-1 approximant, a move of distance 2^{-1}, and an individual can
// neither reach it by deepening nor afford it if it could. It changes across
// generations, by crossover, and that is where a model-backed mutation
// operator earns its place (Sec. 10.2).
| contract Revise(@rung, @refutes, @confirms, ret) = {
  match refutes * 4 > confirms {
    true => { ret!(rung + 1) }
    false => { ret!(rung) }
  }
}

// =====
// THE CHECKER -- Sec. 11.6. An instrument the player HOLDS, sited in the
// world so that sixteen agents share one cache. It receives only on the
// deposited families and NEVER on occ: otherwise the pride would move the
// zebra by looking at it. Verdicts are Kleene {f < b < t} with b carrying
// both readings -- the environment is stuck, or the observer could not
// afford to decide -- and the caller must not be able to tell which.
// =====

```

```

| contract Check(@formula, @posHandle, @algebra, payer, ret) = {
  new r in {
    Live!(*payer, Price(formula), *r)
    | for (_ <- r) {
      Evaluate!(formula, posHandle, algebra, *ret)
      // ret!((verdict, value, witness, cost))
    }
    // If Live cannot fund the query the piece receives nothing, which the
    // protocol reads as verdict "b". Two bottoms, one value, by design.
  }
}

// =====
// THE ARENA -- the rules, holding no stack of its own. It mints nothing;
// it ROUTES. An exogenous reward would need a referee with a mint, and a
// referee with a mint destroys conservation.
//
// It is also where the one unavoidable universal lives. Check, pin and
// discovered check constrain the legality of EVERY piece's move; no piece
// can establish that locally and no quorum can establish it either, because
// an illegal move is not a move and a vote does not make it one. So the
// world refuses, and the player's own hypothesis language stays existential
// and finitely confirmable.
// =====

| contract Turn(b, @ids, @maxR, @who, over) = {
  new chosen in {
    Round!(*b, ids, 1, maxR, *chosen)
    | for (@res <- chosen) {
      new played in {
        Ply!(*b, Move(res), *played)
        | for (@out <- played) {
          match out {
            ("played", m) => {
              Adjudicate!(*b, *over, who)
              | Turn!(*b, Live_ids(ids), maxR, 3 - who, *over)
            }
            // A refused guard left the messages in the tuple space, so
            // the position is untouched and the clock simply retries.
            ("refused", m) => { Turn!(*b, ids, maxR, who, *over) }
          }
        }
      }
    }
  }
}

//! ideal -- mate, stalemate and the draw conditions are modal formulae over
// the deposited families, not scans of a list. The fifty-move and repetition
// rules are the only genuinely HISTORICAL facts in chess and nothing here
// uses the history monad; that is open problem Q4.

| contract Adjudicate(b, over, @who) = {
  new r in {
    @[*b, "state"]!(*r)
    | for (@pos <- r where pos |= @{ Mate(3 - who) }) {
      over!("won", who)
    }
    | for (@pos <- r where pos |= @{ Stalemate(3 - who) }) {
      over!("draw")
    }
  }
}

```

```

| Boot! (Nil)
}

```

B The formula catalogue

Structural predicates, all over deposited families, all at perception prices:

$$\begin{aligned}
\text{Occ}(s, p) &:= \text{out}(\text{occ}(s)) @ (*p) \\
\text{Emp}(s) &:= \text{out}(\text{occ}(s)) @ \mathbf{0} \\
\text{Bears}(p, s) &:= \exists c. \text{out}(\text{atk}(s, c, p)) \top \\
\text{Attacked}(s, c) &:= \exists p. \text{Bears}(p, s) \wedge \text{Colour}(p, c) \\
\text{Defended}(s) &:= \text{Attacked}(s, me) \\
\text{Pinned}(p) &:= \exists s. \text{Occ}(s, p) \wedge \text{out}(\text{pin}(s)) \top \\
\text{Available}(p) &:= \neg \text{Pinned}(p) \wedge \neg \text{Committed}(p) \\
\text{InCheck}(c) &:= \text{out}(\text{chk}(c)) \top
\end{aligned}$$

Derived formulae:

$$\begin{aligned}
\text{Threat}(p) &:= \exists s. \text{Bears}(p, s) \wedge \text{Valuable}(s) \\
\text{Fork}(p) &:= \text{Threat}(p) \mid_0^{N_{atk}} \text{Threat}(p) \\
\text{Hangs}(m) &:= \text{Attacked}(to(m), them) \wedge \neg \text{Defended}(to(m)) \\
\text{Bearing}(s) &:= \langle E \rangle_{N_{pl}}^V \lambda p. \text{Bears}(p, s) \otimes \text{Available}(p) \\
\text{Coherent} &:= \nu X. \langle E \rangle_{N_{pl}}^V ([\beta] \otimes \langle K \rangle X)
\end{aligned}$$

Coherent is the society's analogue of herd coherence: the coalition keeps agreeing as the population steps. It is not on the ladder because it is a ν over the run tree and therefore a depth engine, priced accordingly; it is listed because it is the formula an *observer* of the society would use, and the society is going to be observed.